Tan Yu Min

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Professional Summary

Computer Science undergraduate at the Singapore University of Technology and Design with experience building full-stack web applications using React, Typescript, Go, and modern web technologies. Experienced in leading student engineer teams, developing production software, and building developer-focused tooling with an interest in scalable systems and developer experience.

Core Skills

- Languages: Javascript/TypeScript, Go, C++, C#, Python, Java, SQL
- Full-stack: Next.js, ASP.NET
- Frontend: React, HTML, CSS, Elm
- Backend: Express, Gorilla Mux
- Infrastructure: Docker, Linux, Git, AWS, Supabase, Vercel
- Test Frameworks: Jest, Vitest
- Other Interests: Keyboards, Coffee, Basketball

Professional Experience

Sign Pte Ltd Singapore
Software Engineer Intern March 2021 - August 2021

- Worked with a Linux development environment, using the CLI to install packages to set up the dev environment, and editing files using Vim.
- Gathered and clarified requirements from superiors to ensure that features implemented met the user and business needs.
- On the frontend:
  - Implemented features on Elm frontend, using functional programming paradigms like recursion and higher order functions.
  - Implemented responsive UI, ensuring adequate margins/padding, responsive layouts, as well as a design consistent with the rest of the application for various viewports.
  - Implemented new UX features such as loading skeletons to improve perceived application responsiveness.
- On the backend:
  - Wrote and updated RESTful API endpoints using Express
  - Developed passing and failing Jest unit tests to ensure proper API functionality before deployment.
- Passed strict code review sessions with supervisors before deployment, ensuring principles such as SRP, YAGNI and KISS were upheld to the team's standards.

Academic Projects

SUTD ROOT Student Government root.sutd.edu.sg
Technical Lead / Secretary of ROOTech Mar 2026 - Present

- Currently leading a team of 10 developers to build and maintain internal and external applications for ROOT Student Government and the general student population respectively.
- Liaise with key stakeholders such as ROOT Cabinet and School Offices to gather user/business requirements.
- Plan and assign tasks to developers, and assist with any blockages during development process.
- Review and merge developers' code into production, and request changes if necessary.
- Improved developer workflows and reduced infrastructure costs by evaluating and introducing tools such as SOPS for secret management and migrating backend services from DigitalOcean to Supabase.
- Wrote new documentation and cleaned up old documentation to facilitate the onboarding process for new developers.

ETI Microservice Assignment *Docker, Docker Compose, SSH, PuTTY, Microservices*

- Built the ClassExchange Microservice and Microfrontend as part of a network of 20+ microservices
- Liaised with developers of Class and Authentication Microservices and consumed their APIs to allow for class exchange functionality.
- Built Backend RESTful API with Go and Gorilla Mux, and used MySQL to store class exchange data.
- Containerised Applications with Docker, and used Docker Compose to orchestrate the microservices, allowing for easy deployment and testing.
- Used SSH and PuTTY to connect to the remote server, and successfully deployed the microservices onto the server for testing.

LeafMeBe *ESP32 Microcontroller, Arduino, C++*

- Developed automatic plant-watering system using an ESP32 microcontroller and Arduino ESP Core.
- Interfaced moisture sensors and water pumps via GPIO control to automate watering cycles and eliminate manual watering
- Awarded **Top in Cohort for Innovativeness and Risk-Taking**

Personal Projects

CIS Youth Hymnal *Hugo, HTML, CSS, Javascript* github.com/church-sg/cis-youth-hymnal-v2

- Open-source Progressive Web App used by 50+ church members to browse hymn lyrics with offline support.
- Built with Hugo SSG and served by GitHub Pages for fast static page loads, while Service Workers provide offline functionality.
- Wrote custom JavaScript and CSS to enable features not built into Hugo, e.g.
  - Implemented custom navigation with HTMX, allowing for seamless navigation between pages by replacing full-page browser reloads with partial DOM updates.
  - Large "previous" and "next" buttons for ergonomic browsing on mobile
  - Auto-scrolling to prior location upon returning to list view to reduce repetitive scrolling
  - Page Indexing powered by Lunr to allow for searching both title and content, increasing search accuracy and speed.
- Wrote a Bash script for a 1-click build-and-deploy onto GitHub Pages
- Wrote documentation to onboard new developers quickly to add new hymns using Markdown.

Dotfiles *Bash, Lua* https://github.com/notyumin/dotfiles/

- Personal Linux development environment managed with GNU Stow, including:
  - *GNU Stow*: Allows 1-command symlink of all config files to the right locations, enabling quick, reproducible workstation setup
  - *Neovim*: Built on the MiniMax distro, with custom keybindings and plugins to improve productivity and ergonomics
  - *Tmux*: Custom keybindings and minimal status bar
  - *Wezterm*: Custom colourscheme and UI tweaks

Education

Singapore University of Technology and Design Singapore
BEng. Computer Science 2024 - 2028

School of InfoComm Technology - Ngee Ann Polytechnic Singapore
Diploma in Information Technology 2019 - 2022

School of Science and Technology, Singapore Singapore
O-Levels 2015 - 2018

Awards

- SUTD Freshmore Honours List (2024/2025)
- LeafMeBe: Top in Cohort for Innovativeness and Risk-taking (2025)